Product, Quotient, and Trigonometric Derivatives

Section 1: The Product Rule

1 Find the derivative of $f(x) = x^3 \sin x$.

a. $3x^2 \sin x + x^3 \cos x$

b. $3x^2 \cos x$

c. $3x^2 \sin x - x^3 \cos x$

d. $x^2(3\sin x + \cos x)$

2 If $h(x) = f(x)g(x)$ with $f(2) = 3$, $f'(2) = -1$, $g(2) = 4$, and $g'(2) = 5$, what is $h'(2)$?

a. 11

b. 19

c. -14

d. 15

3 Find $\frac{d}{dx}[e^x \cos x]$.

a. $e^x(\cos x - \sin x)$

b. $e^x(\cos x + \sin x)$

c. $-e^x \sin x$

d. $e^x \cos x$

Section 2: The Quotient Rule

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4 Differentiate $y = \frac{2x-3}{x+4}$.

a. $\frac{11}{(x+4)^2}$

b. $\frac{5}{(x+4)^2}$

c. $\frac{2x+5}{(x+4)^2}$

d. $\frac{2}{(x+4)^2}$

5 Let $k(x) = \frac{f(x)}{g(x)}$. If $f(1) = 6, f'(1) = 2, g(1) = 3,$ and $g'(1) = -1$, find $k'(1)$.

a. $\frac{4}{3}$

b. 0

c. 2

d. $-\frac{4}{3}$

6 Find the derivative of $f(x) = \frac{e^x}{x^2}$.

a. $\frac{e^x(x-2)}{x^3}$

b. $\frac{e^x(x+2)}{x^3}$

c. $\frac{e^x}{2x}$

d. $\frac{e^x(x-1)}{x^3}$

Section 3: Derivatives of Basic Trigonometric Functions

7 What is $\frac{d}{dx}[\tan x]$?

a. $\sec^2 x$

b. $\sec x \tan x$

c. $-\csc^2 x$

d. $\cot x$

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8 What is $\frac{d}{dx}[\sec x]$?

a. $\sec x \tan x$

b. $\sec^2 x$

c. $-\csc x \cot x$

d. $\tan^2 x$

9 What is $\frac{d}{dx}[\cot x]$?

a. $-\csc^2 x$

b. $\csc^2 x$

c. $-\sec^2 x$

d. $-\csc x \cot x$

Section 4: Higher-Order Derivatives & Acceleration

10 Find the second derivative $f''(x)$ of $f(x) = \sin x$.

a. $-\sin x$

b. $\sin x$

c. $-\cos x$

d. $\cos x$

11 If position is $s(t) = t^4 - 2t^3 + 5$, what is the acceleration $a(t)$ at $t = 2$?

a. 24

b. 36

c. 12

d. 48

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12 What is the 50th derivative of $f(x) = \cos x$?

a. $-\cos x$

b. $\cos x$

c. $\sin x$

d. $-\sin x$

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